Bias, Omission, and Control: How Censorship Shapes LLM Responses

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Abstract

Large language models (LLMs) are increasingly being integrated into everyday technologies, and the integrity and diversity of the data they are trained on have become critical public concerns. Online information is not uniformly accessible as governments, organizations, and ISPs employ content filtering, website blocking, and other censorship methods to restrict access to specific online content. Consequences of censorship on society and humans are generally well documented. However, its effects on LLMs that rely on large-scale internet data for training remain less understood.

Unlike human-facing censorship, where blocked websites or restricted content signal that information is being withheld, LLM censorship operates silently and leaves no trace. An LLM that never encountered certain information during training, cannot tell us what it doesn't know, and users also have no way to detect the absence. This creates a problem where we are building AI systems that don't just reflect censorship but naturalize it. It is important to understand how censorship shapes what LLMs can "see" and "learn" to ensure fairness and robustness of these models, in addition to preserving the free flow of knowledge in artificial intelligence (AI) systems.

1 Introduction

This project investigates how internet censorship techniques: blocking, firewalls, flooding, and disinformation, affect the training data and outputs of LLMs. Censorship limits the model's exposure to certain kinds of online content, which could result in biased or incomplete outputs. For instance, if the environment gives higher rewards to actions that reflect social or institutional biases, the RL agent will treat those actions as optimal. This is dangerous because it can reinforce existing discrimination or social marginalization, amplifying unfair biases or discriminatory behaviors in society.

We will aim to characterize how forms of filtering, specifically firewalls and disinformation, may alter the composition of LLM training. We will then analyze how altered data inputs influence model outputs in relation to information omission, bias, and factual consistency. We will also conduct an experiment to determine whether LLMs, specifically ChatGPT-4, reflects disinformation bias using the LIAR2 Dataset. The LIAR2 Dataset is a collection of statements from internet sources (especially social media) that are labeled from 0 to 5 to represent statements that are least to most true. Our experiment will be asking ChatGPT-4 whether the statements in the LIAR2 Dataset are True or False to examine whether the LLM has been trained on biased data or disinformation circulating on the internet and how that may impact LLM outputs.

2 Related Work

Ahmed and Knockel [8] have examined how online censorship shapes large language models (2024) and Wu et al. [3] studied the infrastructural and technical mechanisms of internet censorship, specifically how China's Great Firewall detects and blocks fully encrypted traffic. Our project will extend these two lines of inquiry: bridging their perspectives and findings, and investigating the resulting implications of model training and data curation. confirm the below paragraph for correct citations and literature used We will review prior literature across several themes to ground our research. To examine the general impact of online censorship on LLMs, we will reference Ahmed and Knockel [8] and Glukhov et al. [5], who show how censorship shapes model outputs and information flow. To understand how censorship restricts data access, we will draw on Hoang et al. [6] and studies on firewalls and WAFs Wu et al. [3], which document technical limits on web content and resulting data gaps. To explore how censored or biased data influences LLM behavior, we will examine studies on bias and omission (Yang and Roberts, Hovy and Prabhumoye, Noels et al.) [12] [7] [10] and on disinformation and data poisoning (Jiang et al., Danet, Maus et al.) [2] [4] [9]. Finally, to contextualize applications of censorship-like techniques for IP protection, we will review research on AI and copyright (Wei et al., Barbera) [11] [1] to understand existing methods and their limitations.

3 Project Preparation

We plan to divide the project into two stages: literature review of types of censorship and their researched impact on LLMs, in addition to a practical study simulating and comparing inputting censored versus uncensored data into commercial LLMs. In the first stage, we will examine the impact of censorship on the input data. We will also need to understand how the biased data is ultimately processed within LLM and then outputted, drawing on the mechanics of how LLMs operate.

The data we will use for this stage draws on our preliminary research, suggesting that firewalls, flooding, and disinformation may have impacts on the outputs of LLMs. Broadly speaking, we will investigate two frameworks of censorship: omitting truthful information or adding false information to model input data. We must gather further data to understand specifics on how censorship impacts input data. From here, we will draw on literature about how LLMs and natural language processing techniques process biased data. We will then investigate specific techniques, such as how word embeddings can encode bias in models [12]. Once we establish ways in how censorship techniques impact input data, we aim to draw from research to discuss LLM processing of biased data. Finally, we will extend our understanding to analyze its impact on users, such as in the case of differing outputs from LLMs about events in Chinese history prompted in traditional Chinese versus simplified Chinese [12].

The second stage will be to take our theoretical understandings and apply them to an experimental study of LLMs (inspired by Ahmed and Knockel's study). For our practical experiment, we will expose LLMs to censorship techniques in a controlled setting and observe how this exposure influences their outputs. First, we will create a control trial by providing LLMs with fabricated "training" prompts. Then, we will provide the model with fabricated censored or restricted information (text that omits sensitive topics or follows typical censorship rules from control trial's training data). After this, we will query the model using a standardized set of prompts before and after the censored exposure. By comparing the outputs, we aim to identify differences such as omitted details, altered phrasing, or other indications that the model reflects the censored input. On average, we would expect each trial to differ in output, and trials to differ across different models. The model output may not differ significantly depending on the extent of the manipulation of the base prompt. There may not be a way to quantify outputs of data, but we plan to ensure as much transparency as possible by recording trial responses in a table to compare the control prompt and variables modified.

4 Disinformation

5 Evaluation

For our evaluation of censorship techniques and their impact on LLMs, we will analyze and then compare each of the different techniques. To guide our evaluation, we have generated questions such as: Are the outputs of LLMs different for each mode of censorship? Is there a certain threshold of input data bias the model can filter? Do the different censorship techniques change the way each model processes the data? While quantitative answers may not exist for some of these questions, comparing different modes of censorship will help provide reference points for our evaluation. We plan to make our comparisons more concrete by researching measurements for the degree of censorship on input data. We can similarly research ways to measure the outputs of LLMs.

As for our experimental study, we will evaluate the results in a quantitative manner by categorizing similar types of behavior in model responses and counting the occurrence within each category. Moreover, we can quantify the degree of censorship applied to our inputs by calculating the percentage of change in training data between different trials. We can qualitatively evaluate our study by analyzing whether it matches previous research—in other words, if it is rational to predict a specific model output given the variables we changed and kept the same.

Finally, we can briefly explore the implications of our study on protecting intellectual property, particularly seeing if specific censorship methods we cited in our research can be repurposed to protect certain online content from being outputted by an LLM.

6 Ethics

Regarding the ethical implications of our project, our models may repeat the same types of bias that exists in censored information because we are testing how censorship may impact LLMs. As an example, they may leave out certain opinions or topics that censorship typically hides. This is especially important for our experiment, where we expose the model to examples of censored information—not actually train it on real-life censored training data. Because this exposure does not fully simulate how a model learns from censored training data, it introduces a limitation: the effects we observe may reflect short-term bias from exposure rather than deeper, structural bias that would appear through full training. This means our results could be partial or misleading if interpreted without caution. We also risk reinforcing the same biases we are trying to study, since the model's outputs may already be influenced by the censored inputs we provide. To address this, we will clearly document our process, interpret results carefully, and emphasize that our findings are exploratory rather than definitive.

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